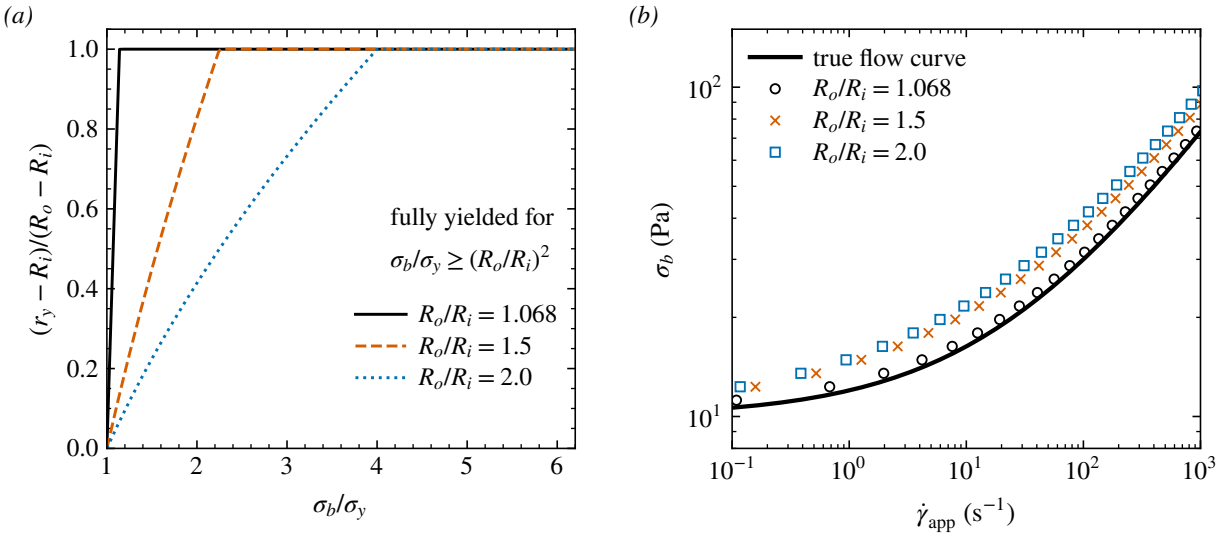
(a)
1.0
0.8
0.6
0.4
0.2
0.0
$(r_y - R_i)/(R_o - R_i)$
1
2
3
4
5
6
σ_b/σ_y
fully yielded for
$\sigma_b/\sigma_y \geq (R_o/R_i)^2$
$R_o/R_i = 1.068$
$R_o/R_i = 1.5$
$R_o/R_i = 2.0$
(b)
true flow curve
$R_o/R_i = 1.068$
$R_o/R_i = 1.5$
$R_o/R_i = 2.0$
10^2
10^1
σ_b (Pa)
10^{-1}
10^0
10^1
10^2
10^3
$\dot{\gamma}_{\mathrm{app}}$ (s^{-1})